The scoring system

A seven-tiered approach will be applied that consists of combining (i) the score of encouraging factors, (ii) the score of the discouraging factors, (iii) standardizing the overall encouraging score, (iv) standardizing the overall encouraging score, (v) computing individual perceived walkability by summing the two standardized scores, (vi) weighting respondents' scores, and (vii) aggregating the score to a geographic area or a demographics group.

First: The scoring of the encouraging factors

The responses of the first section will be combined into a single score through five steps:

- Step (1): For question (i): (Is this factor relevant to your walking needs?), the following condition will be applied:
  - If 'No' was selected, the question will be omitted from scoring.
- Step (2): For question (ii): (Does this factor exist in your neighbourhood?), the following condition will be applied:
  - If 'No' was selected, a score of -6 will be assigned, as part of penalizing but unmet needs.
- Step (3): For question (iii): (To your best knowledge, starting from your home, where do you usually experience this factor?), the scoring uses a multiple discrete function that assigns lower base scores as walking time increases.
  - Up to 10 mins walk: a score of 3 will be assigned.
  - 11-20 mins walk: a score of 2 will be assigned
  - More than 20 mins walk: a score of 1 will be assigned.
- Step (3): The walking time scores are summed together. Example: if the three-walking time were selected, the summed score will be 3+2+1=6.
- Step (4): For question (iv): (How does this factor encourage your walking?), the scoring adjusts for intensity, where selecting 'strongly encourages' or 'somewhat encourages' entails multiplying the walking time score by 1 or 0.5 respectively. Example: if the walking time score (question iii) is 6 and 'somewhat encourages' was selected for intensity, the final score will be (6*0.5) = 3.
- Step (5): The scores for all the encouraging factors are summed together to form one score. The maximum positive score a respondent can get is (21*6) = 126.

Second: The scoring for discouraging factors

The responses of the first section will be combined into a single score through five steps.

- Step (1): For question (i): (Is this factor relevant to your walking needs?), the following condition will be applied:
  - If 'No' was selected, the question will be omitted from scoring.
- Step (2): For question (ii): (Does this factor exist in your neighbourhood?), the following condition will be applied:
  - If 'No' was selected, a score of 6 will be assigned, as part of rewarding the absence of discouraging factors.
- Step (3): For question (iii): (To your best knowledge, starting from your home, where do you usually experience this factor?), the scoring uses a multiple discrete function that assigns lower base scores as walking time increases.
  - Up to 10 mins walk: a score of 3 will be assigned.
  - 11-20 mins walk: a score of 2 will be assigned
  - More than 20 mins walk: a score of 1 will be assigned.
- Step (3): The walking time scores are summed together. Example: if the three-walking time were selected, the summed score will be 3+2+1=6.
- Step (4): For question (iv): (How does this factor discourage your walking?), the scoring adjusts for intensity, where selecting 'strongly discourages' or 'somewhat discourages' entails multiplying the walking time score by -1 or -0.5 respectively. Example: if the walking time score (question iii) is 6 and 'somewhat discourages' was selected for intensity, the final score will be (6) * (-0.5) = -3.
- Step (5): The scores for all the discouraging factors are summed together to form one score. The maximum negative score a respondent can get is (21) *(-6) = 126.

Third: Standardizing the Encouraging Factors Score

Since respondents can opt out of encouraging factors that are irrelevant to them (Question i = "No"), we cannot divide the score of encouraging factors by 21. Instead, we divide the **count of relevant factors** for that specific individual, where "Yes" selected for Question (i).

$$Standardized\ Encouraging\ Score = \frac{(Sum\ of\ Encouraging\ Scores)}{N\ of\ relevant\ encouraging\ factors}$$

This gives us an average score per Relevant encouraging Factors, ranging from -6 to +6.

Fourth: Standardizing the Discouraging Factors Score

Since respondents can opt out of encouraging factors that are irrelevant to them (Question i = "No"), we cannot divide the score of discouraging factors by 21. Instead, we divide the **count of relevant factors** for that specific individual, where "Yes" selected for Question (i).

$$Standardized\ Discouraging\ Score = \frac{(Sum\ of\ Discouraging\ Scores)}{N\ of\ relevant\ discouraging\ factors}$$

This gives us an average score per relevant discouraging Factors, ranging from -6 to +6.

Fifth: Computing Perceived Walkability Score for each respondent (individual)

$$\begin{aligned} Perceived\ Walkability\ Score \\ = (Standardized\ Encouraging\ Score) \\ + (Standardized\ Discouraging\ Score) \end{aligned}$$

-Maximum Positive Scenario (Perfect Walkability): A respondent has all relevant encouraging factors (+6 average) and *no* relevant discouraging features (+6 average): (6 + 6) = 12

-Maximum Negative Scenario (worst walkability): A respondent is missing all relevant encouraging features (-6 average) and *has* all relevant discouraging features (-6 average): (-6) + (-6) = -12

-Neutral Walkability: A respondent has mixed experiences where encouraging factors cancel out discouraging ones: (6) + (-6) = 0

Sixth: Weighting the score of each respondent.

<u>The Weighting Approach:</u> the individual perceived walkability score is weighted (from 0.1 to 1.0) based on the respondent's assessment of the survey's completeness (Part Three).

<u>The Logic: Penalty vs. Reward:</u>

Penalty (Values < 1.0): Lower values are assigned to penalize the outcome when a respondent indicates the survey covered very few factors influencing their walking. This reduces the final score.

Reward (Value = 1.0): A full value of 1.0 is assigned to reward the outcome when the respondent indicates the survey successfully covered all potential factors. This reserves the final score.

Example: Respondent X perceived walkability score is 7 and selected 4 when asked about to what extent does the survey cover potential factors influencing their walking. Accordingly, their final score would be:

- 7 (which is the raw Score) × 0.4 (which is the weight) =2.8 (This is the final weighted score)

Seventh: Computing an aggregate perceived walkability score for a neighbourhood or a demographic profile

The aggregate perceived walkability for a specific neighbourhood or a demographic profile can be computed by averaging the final weighted perceived walkability scores of all respondents within that group.

$$Aggregate\ Perceived\ Walkability\ Score = \frac{(Sum\ of\ Individual\ Weighted\ Scores)}{Total\ number\ of\ individuals}$$